

Homework #6

Hamming Codes and the Information Bit Error Rate

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw6.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw6.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW6.1 *Reading the proof of the channel coding theorem.* Read the proof of the channel capacity theorem in Chapter 7 of Cover & Thomas.

- Write the channel coding theorem. In plain English, succinctly specify what needs to be proved in the forward part of the proof. Do the same with the backward part.
- What is a random code?
- What are the items both the sender and the receiver know? What is the part the receiver does not know?
- We used a maximum-likelihood decision decoder in our information bit error rate analysis. What is the decoder used in the proof of the channel coding theorem?
- What is the key method to prove the backward part?

Problem HW6.2 *The (15, 11) Hamming code.* Every binary Hamming code has a parity-check matrix whose columns are the distinct nonzero binary vectors of length m ; the case $m = 4$ gives the (15, 11) code of rate 11/15.

- Construct a parity-check matrix $H \in \{0, 1\}^{4 \times 15}$ in systematic form $H = [P^T \mid I_4]$, and write the corresponding generator matrix $G = [I_{11} \mid P]$.
- Show that the minimum distance is 3, so that every single error is corrected.
- Show that the code is *perfect*: the $2^4 = 16$ syndromes are exactly accounted for by the no-error pattern and the 15 single-error patterns.
- Over a BSC with crossover probability p , derive the block-error probability of maximum-likelihood decoding, $P_B = 1 - (1 - p)^{15} - 15p(1 - p)^{14}$, and show that its leading term is $\binom{15}{2}p^2 = 105p^2$.
- At $p = 10^{-3}$, compare P_B with that of the (7, 4) code, $1 - (1 - p)^7 - 7p(1 - p)^6 \approx 21p^2$. Which code has the higher rate, and which the lower block-error probability? Discuss the trade this pair illustrates.

Problem HW6.3 *Verifying the weight enumeration.* This chapter's information-BER analysis rests on the counts $N_j(w)$ —the number of weight- w error patterns that carry the received word into the decoding region of the j -th codeword.

- (a) Pick any nonzero codeword $\mathbf{c}_j$ of the (7, 4) code and list the eight words of its decoding region, $\mathbf{c}_j + \mathbf{e}$ for the eight coset leaders $\mathbf{e}$.
- (b) Group the eight words by Hamming weight to recompute the row $N_j(w)$ of the enumeration table, and check it against the table in this chapter.
- (c) Verify the checksum $\sum_j N_j(w) = \binom{7}{w}$ for $w = 2$ and $w = 3$.
- (d) Using the table, re-derive the leading term $P_b \approx 9p^2$ of the information bit error rate, and explain why the exponent is 2 for a single-error-correcting code.

Problem HW6.4 *Error detection vs. error correction.* The weight distribution of the (7, 4) Hamming code is $A_0 = 1$, $A_3 = 7$, $A_4 = 7$, $A_7 = 1$. Suppose the code is used for *error detection only* (as in ARQ): the receiver declares an error unless the received word is itself a codeword.

- (a) Show that an error goes undetected exactly when the error pattern equals a nonzero codeword, so that

$$P_u(p) = \sum_{w>0} A_w p^w (1-p)^{7-w} = 7p^3(1-p)^4 + 7p^4(1-p)^3 + p^7.$$

- (b) Evaluate P_u at $p = 10^{-2}$ and $p = 10^{-3}$, and compare with the information bit error rate P_b of the error-correcting decoder from this chapter at the same p .
- (c) Which failure event is rarer, and by roughly what factor? Discuss when a system designer prefers detection-plus-retransmission (ARQ) to forward error correction (FEC).

Problem HW6.5 *Simulating the (7, 4) code.* Write a MATLAB (or Python) simulation of the (7, 4) Hamming code over the BSC: generate random information words, encode with G , flip each transmitted bit independently with probability p , decode by syndrome lookup, and count information-bit errors.

- (a) Estimate P_b at $p = 10^{-1}$, 3×10^{-2} , 10^{-2} , 3×10^{-3} , 10^{-3} , counting at least 100 bit errors per point.
- (b) Plot the estimates together with this chapter's closed-form P_b on a log-log scale and confirm the slope-2 behavior.
- (c) From your plot, at roughly which p does the (7, 4) code stop being beneficial compared with no coding? Submit your code and the plot with your write-up.

Problem HW6.6 *Typical-set size for the BSC.* In class, we have discussed a little bit about the size of the typical set $2^{nH(Y|X)}$ for a BSC with parameter p .

- (a) The following inequality shall provide insight for that discussion. Show that for any $d \in \{0, 1, 2, \dots, n\}$,

$$\binom{n}{d} \leq 2^{nH(\frac{d}{n})},$$

where $H(\cdot)$ is the binary entropy function. (Hint: use the binomial expansion with parameter $p = d/n$.)

- (b) Find the size of the typical set for $n = 100$ and $\varepsilon = 0.2$ for the BSC with $p = 0.1$. Suppose the use of the typical-set decoder. Determine the set of numbers of channel errors that the decoder can correct. Determine the capacity and the probability of error (a bound is good enough) at the given length.
- (c) Suppose the transmitted codeword is the all-zero sequence, and what is obtained from the channel is a sequence of two 1s and ninety-eight 0s. Is the result jointly typical with the setting given in (b)?